

# Identifying Citi Bike Coverage Gaps

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## 1. Problem Statement

Citi Bike plays an important role in New York City's transportation system, yet access to stations is not evenly distributed across Lower Manhattan. Some blocks and neighborhoods remain underserved despite high population density and strong demand for short-distance mobility.

### Objectives:

- Measure accessibility to Citi Bike stations.
  - Identify areas with limited coverage.
  - Provide a clear, data-driven understanding of where new stations would most effectively improve **mobility equity**.
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## 2. Data Sources & Integration

This project will utilize a multi-layered spatial approach using the following datasets:

- **Citi Bike Station Data:** Sourced from NYC Open Data.
- **Demographic Data:** Census blocks/block groups from U.S. Census TIGER/Line files.
- **Geographic Boundaries:** Neighborhood Tabulation Areas (NTAs) from NYC Planning.

**Technical Note:** All datasets will be imported into **R** using the **sf** package. Layers will be transformed into a common projected Coordinate Reference System (CRS) to ensure accurate distance calculations. The study area will be clipped specifically to Lower Manhattan.

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## 3. Methodology

The analysis will follow a standardized spatial workflow:

1. **Distance Calculation:** Measuring the distance from each census block centroid to the nearest Citi Bike station.
2. **Tiered Classification:** Blocks will be categorized into accessibility tiers (e.g., High, Moderate, Low) based on distance.
3. **Density Mapping:** A point-density surface may be generated to visualize station

concentration.

- 4. **Spatial Joins:** Accessibility metrics will be linked to neighborhood boundaries to facilitate comparative analysis.

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## 4. Analysis Plan

The core of the analysis focuses on quantifying the human impact of infrastructure gaps.

- **Impact Quantification:** Calculating the number of residents living in low-access areas.
- **Gap Identification:** Identifying specific neighborhoods with the largest coverage discrepancies.
- **Suitability Scoring:** (Optional) Combining population density with distance data to highlight "High-Need" locations.
- **Deliverable:** A ranked list of neighborhoods where new stations would yield the greatest benefit.

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## 5. Visualization Goals

High-resolution maps will be the primary output, designed for a general audience:

| Map Type                 | Purpose                                               |
|--------------------------|-------------------------------------------------------|
| Existing Infrastructure  | Displays current Citi Bike station locations.         |
| Accessibility Choropleth | Shows distance-based accessibility by census block.   |
| Coverage Gap Heat Map    | Highlights specific "dead zones" and high-need areas. |

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## 6. Reproducibility

To ensure transparency and future scalability, the project will adhere to the following standards:

- **Environment:** All work conducted in **R** using sf, terra, and tidyverse.
- **Scripting:** Fully documented scripts covering CRS transformations, joins, and analysis.
- **Version Control:** Git will be used to track changes and maintain a transparent workflow.

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## 7. Ethics & Privacy

This project adheres to strict ethical standards:

- **Data Sensitivity:** All datasets are publicly available and contain **no personally identifiable information (PII)**.
- **Anonymization:** Census data is aggregated at the block/neighborhood level.
- **Focus:** The scope is limited to physical infrastructure and neighborhood-level accessibility; no ethical or privacy concerns are anticipated.